

IDS2Eval Scorecard

cic-ids2017-glf

15 ok · 4 warning · 3 flag · 22 checks

FAILED

- **dedup_check**
train duplicates: 0 (0.00%); test rows leaking a train feature-match: 0 (0.00%); test-internal duplicates: 0

ok
- **label_conflict_check**
no feature vector maps to more than one label

ok
- **near_duplicate_class_check**
no near-zero-distance feature vectors found across 13 classes tested

ok
- **leakage_screen**
no single feature or pair dominates importance (top1=15.1%, top2=21.7%)

ok
- **one_rule_check**
single rule 'Source IP <= 1.5' reaches 81.3% test accuracy (83.3% train)

ok
- **feature_auc_ranking_check**
'Bwd Packet Length Max' alone reaches AUC 1.000 identifying 'Heartbleed'; a single feature this separable is worth checking for a leakage artifact

flag
- **identity_column_flag**
standalone AUC by column: {'Source IP': 0.985, 'Destination IP': 0.98, 'Source Port': 0.837, 'Destination Port': 0.931}. Suggest dropping: ['Source IP', 'Destination IP', 'Source Port', 'Destination Port']

flag
- **port_protocol_shortcut_check**
standalone AUC of 'Source Port' + 'Protocol' combined: 0.907. Port and protocol together may predict the label without attack behavior

flag
- **temporal_leakage_check**
no schema.timestamp_column configured

ok
- **temporal_realism_check**
no schema.timestamp_column configured

ok
- **flow_group_leakage_check**
no schema.flow_id_columns configured

ok
- **row_order_leakage_check**
adjacent-row label-transition rate (train=1.00, test=1.00) as a share of what a randomly shuffled ordering would produce

ok
- **homogeneity_test**
12 classes tested; test-to-train and train-internal proximity statistically indistinguishable for every class (consistent with inherent class homogeneity, not train/test leakage)

ok
- **resplit_falsification**
random-split accuracy=0.9995, grouped-split accuracy=0.9994 (drop=+0.0000); grouped split reproduces random-split accuracy, consistent with inherent class homogeneity rather than a split-artifact explanation

ok
- **scenario_holdout_falsification**
no schema.scenario_column configured

ok
- **class_distribution_report**
15 classes, train imbalance ratio (majority:minority) = 201575:1; classes below 1% of train: ['DoS GoldenEye', 'FTP-Patator', 'SSH-Patator', 'DoS slowloris', 'DoS Slowhttptest', 'Bot', 'Web Attack - Brute Force', 'Web Attack - XSS', 'Infiltration', 'Web Attack - Sql Injection', 'Heartbleed']

warning
- **low_cardinality_warning**
unique values by column: {'Source IP': 16328, 'Destination IP': 18766, 'Source Port': 64558, 'Destination Port': 51291}

ok
- **schema_fingerprint_check**
matched extractor signature(s): ['CICFlowMeter']

warning
- **data_integrity_check**
missing values in 1 feature column(s); +-inf values in 2 feature column(s); 8 constant/near-constant feature(s)

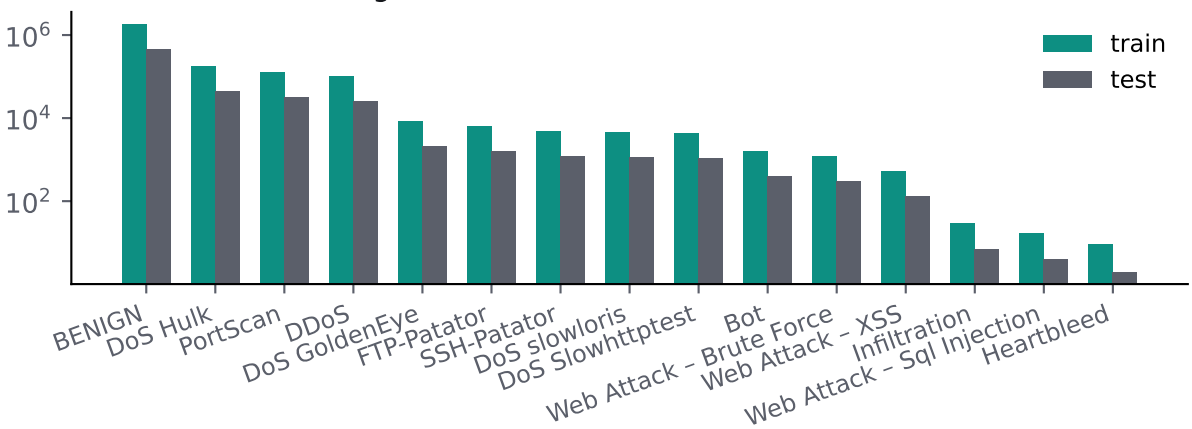
warning
- **result_robustness_check**
accuracy ranges from 0.9976 (identity_columns_dropped) to 0.9995 (random_split) across 4 conditions, a spread of 0.0018

ok
- **known_issue_lookup**
The traffic capture has packet disorder and duplication, and some attacks that were actually launched are not correctly labeled as attack traffic in the released CSVs. (Engelen et al. 2021, IEEE S&P Workshops (WTMC)); An independent re-labeling audit measured a 6.67% overall label corruption rate, with some individual attack classes above 75%; Heartbleed alone is only 11 rows (about 0.022% of the dataset), too few to evaluate reliably regardless of labeling accuracy. (Cantone et al. 2024, IEEE Access)

warning
- **repeated_seed_falsification_check**
grouped-split accuracy drop across 10 seeds: mean=+0.0000, 95% CI=[-0.0000, +0.0001]; not material: even the most generous estimate in the 95% CI (+0.0001) stays below the +0.0100 materiality threshold

ok

Class distribution (log scale)



Top feature importances (leakage_screen)

